

Biology Exam 11: Chapters 1-7 (Full Syllabus), Detailed Marking Report

Seemab Ali | Class 9 | Federal Board (FBISE) | Paper Version 5 | Attempt date: 2026-07-16 | Marked: 2026-07-16

SCORE: 56.4/65 (86.8%, Grade A)

A solid first sitting of a fresh Ch1-7 paper. MCQ 10/12 (below Seemab's usual 11-12/12 baseline), Section B 26.9/33, Section C 19.5/20.

10/12

SECTION A (MCQS)

26.9/33

SECTION B (SHORT)

19.5/20

SECTION C (LONG)

56.4/65

TOTAL, 86.8%

A

GRADE

SECTION A: Multiple Choice Questions (10/12)

#	Question summary	Correct	Seemab	Result	Marks
1	"Biodiversity" refers to	B: Variety of living organisms on Earth	B	Correct	1
2	NOT an importance of biodiversity	C: Formation of fossil fuels	D	Incorrect	0
3	NOT a characteristic of living organisms	C: Crystallization and storage	C	Correct	1
4	Study of diversity + evolutionary relationships	C: Systematics	C	Correct	1
5	Correct statement about cytoplasm	A: 90% water; site of biochemical reactions	C	Incorrect	0
6	Gain in functionality at a higher organization level	C: Emergent properties	C	Correct	1
7	Benedict's test detects	C: Reducing sugars	C	Correct	1
8	Protein supporting skin, bones, cartilage, tendons	B: Collagen	B	Correct	1
9	Actin and myosin, present in muscle cells	B: Muscle contraction/relaxation	B	Correct	1
10	RNA-based enzymes are called	C: Ribozymes	C	Correct	1
11	Wheat/fish/rice production and export value study	B: Bio-economics	B	Correct	1
12	Interphase II (between Meiosis I and II)	B: Involves no further DNA replication	B	Correct	1

Section A: 10/12. Bubble panel was re-verified at 400 DPI zoom before finalising (below Seemab's 11-12/12 baseline) — both misses confirmed as genuine, not a scan misread. Q.2: chose D (formation of fossil fuels IS one of the four options but is the one NOT listed as a biodiversity benefit) instead of C. Q.5: chose C instead of A (cytoplasm is 90% water and the site of biochemical reactions, not the description given in option C).

SECTION B: Short Questions (26.9/33)

Q.2(i), OR: Why are liver cells "most metabolically active"? Two functions.**2/3**

"Liver cells are almost round in shape, with a prominent nucleus, without cytoplasmic organelles. This makes them the most metabolically active cells in the body." Two functions: (i) Detoxification of toxic substances (ii) Recycling of old red blood cells — both correct and each worth 1 mark.

The description reverses the key fact: liver cells are metabolically active BECAUSE they have an abundance of cytoplasmic organelles (mitochondria etc. doing the metabolic work), not because they lack them. Writing "without cytoplasmic organelles" states the opposite of the textbook explanation, so the 1 description mark is not awarded. The two functions are both correct and fully credited.

Q.2(ii), MAIN: List the SEVEN characteristics of all living organisms**3/3**

Movement, Respiration, Sensitivity, Excretion, Reproduction, Nutrition, Growth. All seven correct (order does not matter).

Q.2(iii), MAIN: Identify labelled structures A, B, C, D in the animal cell diagram**3/3**

A = Nucleus, B = Mitochondria, C = Golgi Apparatus, D = Ribosomes. All four correct, matching the key exactly.

Q.2(iv), OR: Name any FIVE functions of proteins, one example protein each**2.4/3**

(1) Haemoglobin — transports oxygen and CO₂ in blood. (2) Keratin — hardens hair, nails, feathers, horns, beaks. (3) Histone — attaches to DNA to form chromosomes. (4) Antibodies — provide immunity against germs. All four correct at 0.6 mark each.

Only FOUR functions given where FIVE were required (mark scheme awards 0.6 per correct function/example pair, any five). A fifth valid option would have been actin/myosin (muscle contraction) or fibrinogen (blood clotting) — 0.6 mark lost for the missing fifth entry.

Q.2(v), MAIN: What are ribozymes? How do they differ from typical enzymes?**2.5/3**

"Most enzymes are made up of chains of amino acids to form 3D proteins, known as globular structure of enzymes. A few enzymes are RNA in nature, called ribozymes." (The "(or)" header was written and then scribbled out, confirming the MAIN question was intended.)

Correctly identifies most enzymes as globular proteins and ribozymes as RNA-based, which is the core distinction asked for. Does not extend to the fuller implication given in the key (that catalytic activity is not limited to proteins, since RNA molecules can also fold into catalytic shapes) — 0.5 mark held back for this thinner elaboration.

Q.2(vi), OR: Structure and function of leaf epidermis and the cuticle**1.5/3**

"Upper Epidermis: is a single layer of cells covered by a waxy substance called cutin, which forms the cuticle. This protects the leaf tissues and prevents loss of water." Fully correct and matches the key (1.5 marks).

The lower epidermis half of the question (stomata, guard cells, gas exchange and water vapour evaporation, worth the other 1.5 marks) was not attempted at all — the answer stops after the upper epidermis.

Q.2(vii), OR: Differentiate between a monomer and a polymer (glucose, starch)**3/3**

"A monomer is a single, simple building-block molecule e.g glucose. A polymer is a large, complex molecule built from many repeating monomer units bonded together. E.g starch." Both definitions and both examples correct and complete.

Q.2(viii), OR: Qualitative vs quantitative observations, with an example of each**3/3**

Clean four-row comparison table (subjective/objective method, quality/quantify things, not accurate-and-not-measurable/accurate-and-measurable, five senses/measuring instruments) plus the exact textbook examples: "A leaf is green" (qualitative) and "A length is 8cm" (quantitative). Full marks, more detail than required.

Q.2(ix), MAIN: TWO plant and TWO animal sources of proteins**3/3**

Plant sources: beans, lentils. Animal sources: meats, milk. All four correct.

Q.2(x), OR: The mitotic (M) phase and its two sub-processes**1/3**

"The mitotic phase is the part of cell cycle during which the cell actually divides to form daughter cells." Correct definition of the M phase (1 mark). Then: "3 reasons: development, growth, sexual/asexual reproduction."

The second part of the OR question asks specifically for the M phase's TWO major sub-processes (karyokinesis and cytokinesis, worth 2 marks). Instead, Seemab wrote reasons why cell division is necessary — content that belongs to the MAIN version of this question, not the OR she headed the answer with. The M-phase definition is credited (1 mark); the mismatched second half earns 0 of the 2 marks for sub-processes.

Q.2(xi), OR: What is systematics? How does it differ from taxonomy?**2.5/3**

"Systematics: is the scientific study of the diversity of organisms and their evolutionary relationships." (1.5/1.5, matches key exactly). "Taxonomy: Taxonomy is the branch of biology that is naming, identification and classification of organelles."

Taxonomy classifies and names ORGANISMS, not organelles (organelles are parts of a cell). This word swap changes the meaning of the definition, so 0.5 mark is held back from the taxonomy half; the systematics half is fully correct.

Section B: 26.9/33. Strongest questions were the diagram identification (2iii), the monomer/polymer and qualitative/quantitative comparisons (2vii, 2viii), and the protein sources question (2ix) — all clean 3/3s. The biggest losses were Q.2(x) (wrong content for the OR's second half, a main/OR mix-up) and Q.2(vi) (only answering half of a two-part question).

SECTION C: Long Questions (19.5/20)**Q.3(i), MAIN: Well-labelled diagram of the transverse section of a leaf****5/5**

A clear, well-proportioned diagram showing the layered structure of a leaf cross-section (columnar palisade cells beneath the upper epidermis, irregular spongy mesophyll cells below, a central vascular bundle, and guard cells flanking a stoma at the base), with EIGHT correctly placed labels: upper epidermis, lower epidermis, phloem, xylem, palisade mesophyll, spongy mesophyll, stoma, guard cell.

Well over the required SIX labels (0.5 mark each, capped at 3), and the diagram structure itself closely matches the textbook figure — full marks for both the diagram (2) and the labelling (3).

Q.3(ii), OR: (a) SEVEN characteristics of life. (b) EIGHT taxonomic ranks.**5/5**

(a) Movement, Respiration, Sensitivity, Excretion, Nutrition, Growth, Reproduction — all seven listed correctly (3/3).

(b) Domain, Kingdom, Phylum/division, Class, Order, Family, Genus, Species — all eight ranks, correct order, broadest to narrowest (2/2).

Both parts complete and fully correct; note the answer to part (b) continues onto the next page without a repeated question header, which is expected since it is the second half of the same OR question, not a gap.

Q.3(iii), MAIN: Functions of proteins — collagen/keratin, actin/myosin, haemoglobin, antibodies**4.5/5**

(a) Collagen/Keratin (1.5/1.5): "Collagen is a major component in the structure of skin, bones, cartilage, muscles, tendons, ligaments, providing support and strength. Keratin is the component of hair, nail, feather, horns and beaks, making them hard and protective."

(b) Actin/Myosin (1.5/1.5): "These proteins are present in muscle cells and are involved in the contraction and relaxation of muscles during movement e.g. locomotion, breathing movements, blood pumping by heart, peristaltic contraction of gut and speech movements."

(c) Haemoglobin (1/1): "RBCs contain red pigment haemoglobin, a protein that transports oxygen and carbon dioxide in the body."

(d) Antibodies (0.5/1): "Antibody proteins, produced by white blood cells."

Part (d) names the source of antibodies correctly (produced by WBCs) but never states their actual function — "provide immunity against germs" — which is the specific thing this sub-part was asking for. 0.5 mark held back for the missing function statement.

Q.3(iv), MAIN: (a) Fat-soluble vitamins. (b) Structure of steroids.**5/5**

(a) "Four fat-soluble vitamins mentioned in the textbook are: vitamin A, D, E, K. They are lipids in nature. Vitamin D is synthesized/derived from cholesterol in the body." (2/2)

(b) "Steroids are lipids that do not have fatty acids in their structure. Instead, they have a distinctive 3-ring fused ring structure. Example: Cholesterol, Vitamin D." (3/3)

Both parts match the answer key almost word for word, including the correct named examples.

Section C: 19.5/20, the strongest part of this paper. The leaf diagram, the taxonomic-rank listing, and the fat-soluble-vitamin/steroid question were all perfect; the only loss was a single missing sentence on the antibody function within an otherwise excellent four-part proteins answer.

Deduction Log (reconciles to headline total)

Question	Marks available	Marks lost	Reason	Marks awarded
Section A, Q.2	1	1	Chose D instead of C (fossil fuels question)	0
Section A, Q.5	1	1	Chose C instead of A (cytoplasm question)	0
Section A, Q.1/3/4/6-12	10	0	All correct	10
Q.2(i) OR	3	1	"Without cytoplasmic organelles" reverses the key fact (organelles ARE abundant); both functions correct	2
Q.2(ii)-Q.2(iii)	6	0	Both fully correct	6
Q.2(iv) OR	3	0.6	Only 4 of 5 required protein functions given	2.4
Q.2(v)	3	0.5	Correct core distinction but thinner on the "why it matters" implication	2.5
Q.2(vi) OR	3	1.5	Lower epidermis / stomata / guard cell half of the question not attempted	1.5
Q.2(vii)-Q.2(ix)	9	0	All fully correct	9
Q.2(x) OR	3	2	M-phase definition correct, but "3 reasons" (main-question content) given instead of the OR's required 2 sub-processes (karyokinesis, cytokinesis)	1
Q.2(xi) OR	3	0.5	Taxonomy defined as classifying "organelles" instead of "organisms"	2.5
Q.3(i), Q.3(ii), Q.3(iv)	15	0	All fully correct	15
Q.3(iii)	5	0.5	Antibodies: source correct, but "provide immunity against germs" function never stated	4.5
Total	65	8.6	Reconciles exactly to headline	56.4

Strengths

- Animal cell diagram labelling (nucleus, mitochondria, Golgi, ribosomes) fully correct
- Both comparison-table questions (monomer/polymer, qualitative/quantitative observations) answered with textbook-precision definitions and examples
- Leaf transverse-section diagram: accurate structure, all 8 labels correct, well beyond the required 6
- Eight taxonomic ranks listed in the correct order without a single slip

Areas to Revise

- **Liver cell organelles reversed (Q.2i):** wrote "without cytoplasmic organelles" when the correct fact is the opposite — liver cells have an ABUNDANCE of organelles, which is precisely why they are metabolically active. Re-read this definition before the next attempt.
- **Main/OR mix-up (Q.2x):** the OR header was written but the second half of the answer supplied content from the MAIN question (reasons for cell division) instead of the OR's required sub-processes (karyokinesis and

- Fat-soluble vitamins and steroid structure answered essentially word-for-word against the key, including both named examples (cholesterol, vitamin D)
 - Three of the four Section C long questions scored a clean 5/5
- cytokinesis). Before answering, re-read which version (MAIN or OR) was actually chosen and make sure every line matches that version.

 - **Half of a two-part question skipped (Q.2vi):** upper epidermis answered well, but the lower epidermis / stomata / guard cells half was never attempted — a recurrence of Seemab's most common Biology pattern (stopping after finishing only the first half of a joined question).
 - **Function statement dropped after naming a protein source (Q.3iii-d):** antibodies correctly traced to white blood cells, but the actual function ("provide immunity against germs") was never written down.
 - **Organism/organelle word swap (Q.2xi):** taxonomy defined as classifying "organelles" rather than "organisms" — likely a slip of the pen, but it changes the meaning of the definition.
 - **MCQ:** two misses this attempt (Q.2 biodiversity importance, Q.5 cytoplasm composition) — below the usual 11-12/12 baseline; both confirmed correct on re-read at high zoom, so this is a genuine result, not a scan error.

Comparison vs Previous Biology Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
3 Apr 2026	exam-01-ch1-6	11/12	28/33	13/20	52/65	80.0%	A
30 Apr 2026	exam-02-ch7 (1st)	12/12	31/33	17.5/20	60.5/65	93.1%	A
2 May 2026	exam-02-ch7 (re-sit)	12/12	33/33	20/20	65/65	100%	A+
8 Jun 2026	exam-03-ch4	12/12	31/33	19.5/20	62.5/65	96.2%	A+
23 Jun 2026	exam-07-ch1-7	12/12	27/33	18.5/20	57.5/65	88.5%	A
29 Jun 2026	exam-08-ch1-7	12/12	25/33	17.5/20	54.5/65	83.8%	A
6 Jul 2026	exam-09-ch1-7	12/12	32/33	17.25/20	61.25/65	94.2%	A+
16 Jul 2026	exam-11-ch1-7 (this exam)	10/12	26.9/33	19.5/20	56.4/65	86.8%	A

This is a fresh, previously-unseen full-syllabus Ch1-7 paper (exam-11, Version 5), sitting between exam-08 (83.8%) and exam-07 (88.5%) among the four comprehensive Ch1-7 papers so far. Section C (19.5/20, 97.5%) is Biology's strongest Section C result yet, ahead of exam-09's 17.25/20. The MCQ result (10/12) is Biology's weakest bubble-panel score to date, breaking a 13-exam streak of 12/12 and confirmed genuine (not a scan misread) after a high-zoom re-check. Section B (26.9/33, 81.5%) is below Biology's recent norm, driven mainly by the Q.2(x) main/OR mix-up and the incomplete Q.2(vi) answer.